



Technological Innovation Capabilities and Organizational Growth in Early-Stage Fintech Startups

Rachael Atieno Ondijo^{*ID}, Ann Mwangi, Susan Wasike

School of Business, Catholic University of Eastern Africa, 13567-00800 Nairobi, Kenya

*Correspondence: Rachael Atieno Ondijo (1064157@cuea.edu)

Received: 07-13-2025

Revised: 08-29-2025

Accepted: 09-08-2025

Citation: Ondijo, R. A., Mwangi, A., & Wasike, S., (2025). Technological innovation capabilities and organizational growth in early-stage fintech startups. *J. Organ. Technol. Entrep.*, 3(3), 141–150. <https://doi.org/10.56578/jote030301>.



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Abstract: Technological innovation is central to the development and scaling of fintech startups, yet its relationship with organizational growth remains insufficiently understood among early-stage ventures in emerging markets. This study investigates how technological innovation capabilities are associated with organizational growth in early-stage fintech startups, drawing on the technological dimension of the Technology–Organization–Environment (TOE) framework. Technological innovation was examined through product innovation, scalable digital technologies, systems integration, and cybersecurity. A positivist research approach combining descriptive and explanatory designs was adopted. Data were collected through structured questionnaires from founders, co-founders, and senior managers of early-stage fintech startups in Nairobi City County, Kenya. A total of 104 valid responses were obtained from a sample of 131 startups drawn from a target population of 195 firms. The data were analyzed using descriptive statistics, Pearson’s correlation analysis, and simple linear regression. The results showed a positive and statistically significant relationship between technological innovation and organizational growth (unstandardized regression coefficient $B = 0.43$, standardized regression coefficient $\beta = 0.36$, t -statistic = 3.90, probability value $p < 0.001$). Technological innovation explained 13.0% of the variance in organizational growth, with a coefficient of determination (R^2) of 0.130, indicating that technological capabilities were associated with growth while leaving substantial variation attributable to other organizational and environmental conditions. The findings indicate that technological innovation constitutes an important organizational capability for early-stage fintech startups, particularly through continuous product development, scalable digital infrastructure, systems integration, and cybersecurity. This study provides firm-level evidence linking technological capabilities with entrepreneurial growth and shows how the technological dimension of the TOE framework can be applied to understand organizational outcomes in technology-intensive new ventures.

Keywords: Technological innovation; Organizational growth; Fintech startups; Technological capabilities; Digital entrepreneurship; Technology–Organization–Environment framework

1. Introduction

Technological innovation has become a critical strategic capability in the financial technology (fintech) industry, enabling firms to improve operational efficiency, develop innovative products, enhance customer experience, and compete in rapidly changing digital markets (Gomber et al., 2018; Lee & Shin, 2018). Technologies such as cloud computing, artificial intelligence (AI), application programming interfaces (APIs), data analytics, and cybersecurity have increasingly shaped how financial services are developed and delivered. For early-stage fintech startups, continuous technological innovation may therefore be particularly important for achieving and sustaining organizational growth.

Despite the rapid expansion of fintech ecosystems globally and across Africa, many early-stage startups face difficulties progressing from market entry to sustainable growth. These challenges include limited technological resources, cybersecurity risks, infrastructure constraints, and the need to continuously adapt digital products to changing customer and market demands (GSM Association., 2023; World Bank, 2022). Kenya provides an

important context for examining this relationship because of its well-established digital financial ecosystem and growing fintech sector. However, despite increasing adoption of digital technologies, limited firm-level evidence exists on how technological innovation relates to the organizational growth of early-stage fintech startups in Kenya.

Existing studies have largely examined technological innovation in established organizations or specific technologies, with less attention to technological innovation as a multidimensional capability encompassing product innovation, scalable digital technologies, systems integration, and cybersecurity (Autio et al., 2018; Bharadwaj et al., 2013). The Technology–Organization–Environment (TOE) Framework provides a basis for understanding how technological capabilities interact with organizational and environmental conditions to shape organizational outcomes (Tornatzky & Fleischer, 1990).

Against this background, this study examines the relationship between technological innovation and organizational growth of early-stage fintech startups in Nairobi City County, Kenya. The study specifically considers product innovation, scalable digital technologies, systems integration, and cybersecurity as dimensions of technological innovation and assesses their relationship with organizational growth.

To fill this gap, the study aimed to present firm-level evidence from the fintech ecosystem with a specific interest in early-stage fintech startups that had a minimum running time of 1–6 years in the Kenyan ecosystem. The findings from recent studies showed that recent fintech startups in Kenya are leveraging incremental innovation and technology to create new fintech services and stay competitive (Hakizimana et al., 2023). While previous studies had generally focused on studying innovation, fintech performance, or specific technologies, few studies were found that considered a package of technologies in relation to the survival of fintech startups. Technological innovation was thus conceptualized as a multi-faceted phenomenon, including: Product innovation, Scalable digital technologies, Systems integration, and Cybersecurity. This was because digital innovation and technology adoption had become a critical issue in entrepreneurship, productivity, and creating digital businesses in Africa, as per evidence (African Union Commission & OECD, 2021; Anyangaa et al., 2025). Cybersecurity was also considered a critical technology competency for fintech startups, as they continuously leveraged technology, data, and an integrated financial system, creating and utilizing risks for fintech startups (Samme-Nlar, 2022). The study drew a context-specific finding by considering specifically the fintech firms that had been in operation for 1 and 6 years in Nairobi City County and how technological innovation correlated with organizational growth in this period of critical fintech development. The study, as a result, helped move the investigation of technological innovation and small business development within Kenya from theoretical conceptualization to empirical analysis, and offered practical recommendations to fintech entrepreneurs, investors, policymakers, and other actors aiming to enhance the sustainability and growth of early-stage fintech businesses.

2. Literature Review and Hypothesis Development

2.1 Technological Innovation and Organizational Growth

Tornatzky & Fleischer (1990) developed the TOE Framework, which describes three dimensions of context (technological, organizational, and environmental) representing the context in which technology adoption and implementation take place at the organizational level. Technological context includes the features and availability of the technologies concerned with an organization, and the organizational context includes internal features of the organization like organizational resources, structure, managerial support, and readiness. The environmental context includes all relevant external considerations such as competition, regulations, the market, and the nature of the industry. The framework has been proven to be valid in organizing the different technology adoptions in various kinds of organizational and industrial contexts (Awa et al., 2017; Oliveira & Martins, 2011).

In the fintech sector, the technological context takes on an increasingly salient role, as the delivery of financial services is becoming more reliant on digital technologies and technology-based business models. Fintech companies leverage digital technologies such as digital platforms, cloud computing, AI, APIs, and data analytics to develop financial products, automate financial processes, connect with other actors in the fintech ecosystem, and extend financial services. These technological capabilities can enhance operational efficiency, support product development, and create customer value, thereby strengthening the ability of fintech firms to grow (Hakizimana et al., 2023; Thakor, 2020). Moreover, the effective implementation of fintech solutions can contribute to improved business performance and sustainability, particularly in a rapidly changing financial services landscape (Taneja et al., 2024).

In the current study, emphasis was placed on the technological dimension of the TOE Framework since technological innovation was the main explanatory factor being analyzed. This emphasis was especially pertinent to early-stage fintech startups, which are more likely to have a technology-based business and whose success and competitiveness in growing could strongly rely on their ability to develop, adopt, integrate, and secure digital technology. Product innovations can enable the new or enhanced delivery of financial services and digital technologies that can be scaled up and enable services to be expanded without extra physical infrastructure investments. The interoperability of systems through systems integration can help financial institutions and other

participants in the ecosystem, and cybersecurity can guarantee safety against technological threats to the digital systems, customer information, and financial transactions (Hakizimana et al., 2023; Thakor, 2020).

Technological innovation was therefore investigated based on four dimensions: product innovation, scalable digital technologies, systems integration, and cybersecurity. Product innovation was defined as the process of creating and enhancing fintech products and services in order to meet the needs of customers and the market. Scalable digital technologies were defined as the implementation of technologies that could sustain higher transaction volume, customer count, and geographical expansion. Systems integration was the term used to describe the interoperability and connectivity of technological systems within the firm and between the firm and external systems. Cybersecurity focused on the technology solutions implemented to protect information systems, customer data, and digital financial transactions. These dimensions were selected after considering the technological aspects for building digital financial enterprises and the size of the enterprises (Anyangaa et al., 2025; Hakizimana et al., 2023; Thakor, 2020).

The “organizational contexts” and “environment” were ignored as independent variables in this study. This delimitation followed the specific objective of this study, which is to analyze how technological innovation is associated with organizational growth, and not to test the full range of determinants suggested by the TOE Framework. Organizational factors such as managerial support, organizational structure, financial resources, and employee capabilities can impact the adoption and effective use of technological innovations. Likewise, external factors, such as competition, regulations, customer and market conditions, can impact the performance and technological decisions made by fintech companies (Awa et al., 2017; Oliveira & Martins, 2011). Their omission, therefore, was a conscious focus on a limited scope of analysis, more than an unwarranted exclusion of the factors due to their irrelevance to the process of technological innovation or to the expansion of business organizations.

The TOE Framework was thus used not as a three-part framework to be tested within the same study, but as a theoretical lens to apply to the technological context and understand the relevance for the organizations’ outcome. Hence, this study was not aimed at in-depth empirical testing of the TOE Framework. Instead, it addressed the question of what role technological innovation, defined as product innovation, scalable digital technologies, system integration, and cybersecurity, plays in the growth of early-stage fintech startups. This application ensured the consistency of the theoretical framework, variables of the study, and also the objectives of the study.

Empirical evidence additionally affirms the relevance of technological innovation to organizational outcomes in financial and entrepreneurial settings. Based on the literature review, Thakor (2020) found that fintech innovation can serve to decrease transaction costs, finance intermediation, and financial service delivery efficiency. The results of this research follow the study undertaken by Hakizimana et al. (2023), showing the strategic importance of technological innovation in the development of fintech startups in Kenya in an environment of fast technological change and competition. Anyangaa et al. (2025) also shared their finding on the role of fintech in entrepreneurial outcomes in Kenya. Despite the fact that it was not a direct empirical test of the entire TOE Framework, these findings did provide some empirical support to investigate technological innovation as being a possible determinant of organizational growth.

Based on the technological dimension of the TOE Framework and the empirical evidence examined, the study analysed the relationship between technological innovation and organizational growth of early-stage fintech start-up businesses in Nairobi City County in Kenya. The resource-centric theory was developed on the basis that technological capabilities would facilitate the conditions for fintech startups to innovate products and processes, satisfy service delivery, extend customer reach, and increase their scaling capacity. The study therefore hypothesized that Technological Innovation is significantly related to organizational growth of early-stage fintech startups in Nairobi City County, Kenya.

The conceptual framework illustrating the relationship between the study variables is presented in Figure 1. As shown in Figure 1, technological innovation is the independent variable, while organizational growth is the dependent variable.

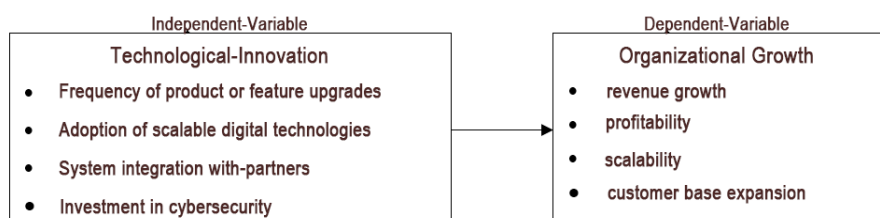


Figure 1. Conceptual framework

Technological innovation is measured using indicators adapted from previous studies, including the frequency of product or feature updates, the adoption of scalable digital technologies, system integration with strategic partners, and investment in cybersecurity. These technological capabilities are expected to enhance operational

efficiency, foster innovation, strengthen customer trust, and expand market access (Hakizimana et al., 2023; Thakor, 2020).

Organizational growth is measured using indicators such as revenue growth, profitability, scalability, and customer growth. The framework therefore posits that stronger technological innovation capabilities are associated with improved organizational growth among early-stage fintech startups in Nairobi City County, Kenya (Anyangaa et al., 2025).

H₁: Technological innovation has a positive and statistically significant relationship with organizational growth among early-stage fintech startups in Nairobi City County, Kenya.

3. Methodology

3.1 Research Design

The study adopted a positivist research philosophy and employed descriptive and explanatory research designs. The positivist philosophy was appropriate because the study sought to objectively examine the relationship between technological innovation and organizational growth using quantitative data and statistical analysis. The descriptive design facilitated the description of the characteristics of the sampled fintech startups, while the explanatory design enabled the study to examine the relationship between technological innovation and organizational growth.

3.2 Study Area and Population

The study was conducted among early-stage fintech startups operating in Nairobi City County, Kenya. Nairobi provides an important setting for the study because it concentrates technology-enabled financial businesses and its position as a major fintech hub in Kenya.

The target population comprised 195 active early-stage fintech startups operating in Nairobi City County. The firms had been in operation for between one and six years and were actively providing technology-enabled financial services. The population was identified using information from the Central Bank of Kenya (2025) and the Kenya National Innovation Agency (2024).

3.3 Sample Size and Sampling Procedure

The sample size was determined using Yamane (1967)'s formula:

$$n = N / [1 + N(e^2)]$$

where, n represents the required sample size, N represents the target population, and e represents the level of precision.

Using a target population of 195 firms and a 5% level of precision, a sample of 131 fintech startups was obtained. One key informant was selected from each sampled startup. Founders, co-founders, and senior managers were purposively selected because of their involvement in strategic decision-making and their knowledge of their firms' technological strategies and organizational performance. A total of 104 completed questionnaires were returned, representing a response rate of 79.39%.

3.4 Data Collection

Primary data were collected using a structured questionnaire administered electronically through Google Forms. The questionnaire comprised closed-ended items measured on a five-point Likert scale, ranging from 1 = strongly disagree to 5 = strongly agree.

Technological innovation was operationalized using four dimensions: continuous product innovation, adoption of scalable digital technologies, systems integration, and investment in cybersecurity. Organizational growth was measured using indicators of revenue growth, profitability, customer base expansion, and business scalability.

Participation was voluntary, and informed consent was obtained from respondents before completion of the questionnaire. Only questionnaires with at least 70% completion were included in the analysis.

3.5 Validity and Reliability

Content validity was established through expert review by the university supervisors and by aligning the questionnaire items with measures reported in previous studies. Reliability was assessed using Cronbach's alpha coefficient. The technological innovation construct recorded a Cronbach's alpha of 0.878, while organizational

growth recorded 0.869, indicating satisfactory internal consistency for both constructs.

3.6 Data Analysis

The collected data were coded and analysed using the Statistical Package for the Social Sciences, version 25. Descriptive statistics, including means and standard deviations, were used to summarize the levels of technological innovation and organizational growth. Pearson's product-moment correlation analysis was used to examine the direction and strength of the relationship between technological innovation and organizational growth.

Simple linear regression analysis was subsequently conducted to determine whether technological innovation significantly predicted organizational growth. Since the study examined one independent variable, namely technological innovation, and one dependent variable, namely organizational growth, a simple linear regression model was considered appropriate. Statistical significance was assessed at the 5% significance level, where p represents the probability value ($p < 0.05$).

The regression model was specified as:

$$OG = \beta_0 + \beta_1 TI + \varepsilon$$

where, OG represents organizational growth, β_0 represents the regression constant, β_1 represents the regression coefficient for technological innovation, TI represents technological innovation, and ε represents the error term.

3.7 Ethical Considerations

Ethical approval for the study was obtained from the Catholic University of Eastern Africa, while permission to conduct the research was obtained from the National Commission for Science, Technology and Innovation. Participation was voluntary, and informed consent was obtained from all respondents. Confidentiality and anonymity were maintained throughout the study, and the data collected were used exclusively for academic purposes.

4. Results

4.1 Response Rate and Sample Characteristics

A total of 131 questionnaires were administered to early-stage fintech startups in Nairobi City County, of which 104 were fully completed and considered suitable for analysis, representing a response rate of 79.39%. The respondents comprised founders, co-founders, and senior managers who were considered sufficiently knowledgeable about their firms' operations, technological strategies, and organizational performance.

4.2 Reliability and Descriptive Statistics

The internal consistency of the study variables was assessed using Cronbach's alpha. The reliability coefficients for all study constructs exceeded the recommended threshold of 0.70. Technological Innovation recorded a Cronbach's alpha of 0.878, while Organizational Growth recorded 0.869, indicating satisfactory internal consistency.

As presented in Table 1, the results indicate that respondents generally reported relatively high levels of technological innovation and organizational growth among early-stage fintech startups in Nairobi City County, Kenya. The standard deviation (SD) indicates the dispersion of responses around the mean. Technological innovation recorded a mean score of 4.16 (SD = 0.51), while organizational growth recorded a mean score of 3.85 (SD = 0.61). These findings suggest that the sampled fintech startups exhibited relatively strong technological innovation capabilities alongside moderate to high levels of organizational growth.

Table 1: Reliability and descriptive statistics

Variable	Cronbach's α	M	SD
Technological innovation	0.878	4.16	0.51
Organizational growth	0.869	3.85	0.61

Note: $n = 104$. M = mean; SD = standard deviation. Variables were measured on a five-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). Cronbach's α values above 0.70 indicate acceptable internal consistency.

4.3 Diagnostic Tests for Simple Linear Regression

As presented in Table 2, the assumptions underlying simple linear regression were assessed before conducting

the regression analysis to confirm the suitability of the data. The diagnostic tests examined the normality of residuals, homoscedasticity, independence of errors, and multicollinearity where applicable to determine whether the explanatory variables were excessively correlated.

Table 2. Diagnostic test results

Assumption	Test/Assessment	Result	Decision
Linearity	Scatterplot of technological innovation and organizational growth	Approximately linear	Satisfied
Normality	Shapiro–Wilk test + histogram/P–P Plot	$p = 0.214$	Satisfied
Homoscedasticity	Residuals vs. predicted values scatterplot	Random dispersion	Satisfied
Independence of errors	Durbin–Watson test	1.89	Satisfied

Note: P–P Plot = probability–probability plot; p = probability value.

The results indicated that the assumptions for multiple linear regression were reasonably satisfied. The Shapiro–Wilk test and the Normal P–P Plot (Probability–probability plot) confirmed that the residuals were approximately normally distributed. In addition, the scatter plot of standardized residuals against standardized predicted values showed a random and relatively even dispersion of residuals, indicating that the assumption of homoscedasticity was met. The relationship between technological innovation and organizational growth was also found to be approximately linear, while the Durbin–Watson statistic of 1.89 indicated that the residuals were independent and that no significant autocorrelation was present.

Overall, the diagnostic results presented in Table 2 confirmed that the dataset met the key assumptions required for simple linear regression analysis, supporting the use of the regression model to examine the relationship between technological innovation and organizational growth among early-stage fintech startups in Nairobi City County.

4.4 Correlation Analysis

Pearson’s product-moment correlation analysis was conducted to establish the relationship between technological innovation and organizational growth. As presented in Table 3, the results revealed a positive and statistically significant relationship between technological innovation and organizational growth, pearson correlation coefficient (r) = 0.36, $p < 0.05$. This finding indicates that higher levels of technological innovation were associated with higher levels of organizational growth among early-stage fintech startups in Nairobi City County, Kenya.

Table 3. Correlation between technological innovation and organizational growth

Variable	Technological Innovation	Organizational Growth
Technological innovation	1.00	0.36*
Organizational growth	0.36*	1.00

Note: $n = 104$; r = Pearson correlation coefficient; p = probability value.

* Correlation is significant at the 0.05 level (two-tailed).

4.5 Regression Analysis

Simple linear regression analysis was conducted to determine whether technological innovation significantly predicted organizational growth among early-stage fintech startups in Nairobi City County, Kenya. Technological innovation was entered as the independent variable, while organizational growth was specified as the dependent variable.

The regression results were evaluated using the coefficient of determination (R^2), the unstandardized regression coefficient (B), the standardized regression coefficient (β), the t -statistic, the F -statistic (F), and the corresponding probability value (p). A p -value below 0.05 was considered evidence of a statistically significant predictive relationship between technological innovation and organizational growth.

Table 4. Simple linear regression results for technological innovation and organizational growth

Predictor	B	SE	β	t	p
Constant	2.06	0.44	-	4.68	< 0.001
Technological innovation	0.43	0.11	0.36	3.90	< 0.001

Note. $n = 104$; Dependent variable: organizational growth; Predictor: technological innovation; B = unstandardized regression coefficient;

SE = standard error; β = standardized regression coefficient; t = t -statistic; p = probability value. “-” indicates that the standardized coefficient is not computed for the constant term. Model summary: $R = 0.36$; $R^2 = 0.130$; adjusted $R^2 = 0.121$; $F(1, 102) = 15.19$; $p < 0.001$.

As presented in Table 4, technological innovation was a positive and statistically significant predictor of organizational growth ($B = 0.43$, $\beta = 0.36$, $t = 3.90$, $p < 0.05$). The model explained 13.0% of the variance in organizational growth ($R^2 = 0.130$). These findings indicate that higher levels of technological innovation were associated with higher levels of organizational growth among early-stage fintech startups in Nairobi City County, Kenya.

4.6 Hypothesis Testing

The study hypothesized that technological innovation has a positive and statistically significant relationship with organizational growth among early-stage fintech startups in Nairobi City County, Kenya.

The regression results showed that technological innovation was a positive and statistically significant predictor of organizational growth ($B = 0.43$, $\beta = 0.36$, $t = 3.90$, $p < 0.05$). The null hypothesis was therefore rejected, and the alternative hypothesis was supported.

The findings provide empirical support for the hypothesis that technological innovation significantly predicts organizational growth among early-stage fintech startups in Nairobi City County. Given the cross-sectional research design, however, the finding is interpreted as evidence of a statistically significant predictive relationship rather than definitive proof of causality.

4.7 Discussion of Findings

The study found that there was a positive and statistically significant relationship between technological innovation and organizational growth among early-stage fintech startups in Nairobi City County. Results of Pearson correlation analysis result in forming a moderate positive association between technological innovation and organizational growth ($r = 0.36$, $p < 0.05$), and the results of simple linear regression analysis indicated that technological innovation is a significant positive predictor of organizational growth ($B = 0.43$, $\beta = 0.36$, $t = 3.90$, $p < 0.05$). The model accounted for a substantial amount of variance in organizational growth, with an R^2 value of 0.130, indicating that technological innovation explained 13.0% of the variance in organizational growth among the sampled startups. The result is aligned with the result obtained by Thakor (2020), which concluded that fintech innovation has the potential to improve financial services efficiency by lowering transaction costs and improving financial intermediation. Beyond that, Hakizimana et al. (2023) emphasized the pivotal role technological innovation plays in the development and competitiveness of Fintech Startups in Kenya. Though the same results were obtained across all the tests, it can be suggested that technological capabilities are an essential part of growth in technology-intensive financial enterprises.

The study shows that the finding aligns with the existing literature, which shows that digital capabilities enable innovation, efficiency, scalability, and customer value creation. Working collaboratively, network effects, and access to larger market opportunities can lead to greater entrepreneurial growth, as Nambisan et al. (2019) determined that digital platforms enable this development. In a similar vein, Autio et al. (2018) showed that digital technologies can help businesses grow and improve their scaling up quickly without relying too heavily on other physical infrastructure. Such arguments would apply to early-stage fintech startups in which technological capabilities, such as product and feature enhancements, scalable digital innovations, systems integration, and cybersecurity, are key. This indicates that in this study, technological innovation capability had a positive regression coefficient; hence, startups with technological innovation capability are more likely to achieve higher organizational growth in terms of revenue growth, profitability, scalability and growth of the customer base. The result also corroborates Anyangaa et al. (2025), who also presented evidence on how fintech impacts entrepreneurial outcomes in Kenya. However, the comparatively low R^2 result of 0.130 suggests that technological innovation alone does not explain the majority of the variation in organizational growth and that other organizational and environmental factors may also influence organizational growth. This observation is consistent with the broader challenge of sustaining fintech startups beyond the early stages of development, where technological and innovation strategies may need to evolve as firms mature (Kollie, 2025).

Based on the theoretical point of view, the study's result contributes to the empirical evidence that the technological dimension in the TOE Framework is relevant to the study of fintech startups. Technological conditions are one of the important contexts in which firms adopt and use technologies as postulated by the TOE Framework (Tornatzky & Fleischer, 1990). The substantial positive correlation result found in this study suggests that technological capabilities were correlated with the growth of the organization, so that technological context can be seen as an explanatory variable. However, complete TOE was not used, and the organizational and environmental aspects were not included in the model, so the results cannot be interpreted as proof of the total TOE Framework. The finding therefore adds to existing literature, bringing firm-level information from early-stage fintech startups in Nairobi City County, where the focus on such services or firms has not been well studied empirically. As an across-section study, this data reflects a strong association and not conclusive evidence of the cause-and-effect of technological innovation and organizational growth.

5. Conclusion

The study has therefore found that technological innovation had a positive and significant correlation with organizational growth for Nairobi City County's early-stage fintech start-up firms. Results from the correlation showed a moderate, positive correlation between the two variables ($r = 0.36, p < 0.05$), and the regression results showed that technological innovation significantly predicted organizational growth ($\beta = 0.36, p < 0.05$). Technological innovation accounted for 13.0% of the variability in organizational growth. Consequently, the results showed that the startups who had more positive technological innovation capability also tended to have more positive organizational growth results.

The study also concluded that technological innovation, specifically when considering a product or feature improvement, the presence of scalable digital technology, the integration with systems, and technological security were considered as relevant technological capabilities in the context of early-stage fintech startups. But the results failed to show that the technological innovation was the only factor that led to organizational growth. The R^2 value for this was relatively low; hence, there was a large unexplained variance in the organizational growth variable analyzed in the study. Additionally, the use of a cross-sectional design in the study led to a lack of definitive evidence of causality but merely a strong association and a predictable relationship between selected variables.

6. Recommendations

The results indicated that technological innovation had a positive and significant influence on the organizational growth of early-stage fintech startups, therefore, the management should improve the technological innovation capability. Continuous product and feature enhancements, scalable digital technologies, business partner integration and cybersecurity are highlighted. To see that these areas are clearly aligned with customers' needs and business expansion goals aiming at improving business revenue growth, profitability, scalability, and expansion of the customer base.

Managers of the fintech should also set up proper and systematic systems for reviewing and upgrading the technical ability of fintech. This can involve evaluating the effectiveness of current digital tools, figuring out how to make the product better, increasing systems integration and continuously evaluating cybersecurity needs. These types of programs should be based on quantifiable organizational goals and on resources that are available to the organization instead of the adoption of technology for technology's sake.

Policymakers and fintech ecosystem players should consider measures that ease access to relevant technological and technical capacities for early-stage startups, as well as measures to provide innovation support and digital infrastructure. This could help startups to develop and implement technology solutions. The policy maker and manager, however, should be aware that technological innovation alone explained 13.0% of the variation in the organizational growth, other organizational and environmental factors likely hold possibilities in understanding the growth of start-ups, which should not be overlooked.

7. Limitations and Suggestions for Future Research

The study focused on early-stage fintech startups operating in Nairobi City County, and therefore the findings may not be directly generalizable to fintech startups operating in other counties or countries. In addition, the cross-sectional design captures the relationship between technological innovation and organizational growth at a particular point in time and does not establish long-term causal effects.

The research has been limited to the technological dimension of the TOE Framework, and the organizational and environmental dimensions have not been studied in an empirical sense. As a result, they did not consider, on the one hand, organizational structure, managerial support, financial and human resources, regulatory requirements, competitive pressures, and market conditions as the explanatory variables for the study. The fact that they fail to capture the findings' significance in broader conditions is where their deficiency lies: technological innovation as a means of achieving organizational growth. Hence, these findings were interpreted as evidence of the relationship between technological innovation and the growth of the organization but not as a full empirical test of the whole TOE Framework.

Future studies could employ longitudinal designs to examine how technological innovation affects organizational growth over different stages of a fintech firm's development. Comparative studies involving fintech startups across different regions of Kenya or other emerging markets could also provide further evidence on whether the observed relationship varies across different regulatory, technological, and market environments.

Author Contributions

Conceptualization, R.A.O., A.M., and S.W.; methodology, R.A.O. and A.M.; software, R.A.O.; validation, A.M. and S.W.; formal analysis, R.A.O.; investigation, R.A.O.; resources, R.A.O.; data curation, R.A.O.; writing—

original draft preparation, R.A.O.; writing—review and editing, A.M., S.W., and R.A.O.; visualization, R.A.O.; supervision, A.M. and S.W.; project administration, R.A.O. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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Appendix

An extract of the questionnaire

Please read each statement carefully and tick (✓) the option that best represents your opinion.

Responses are based on a 5-point Likert scale:

1 = Strongly Disagree; 2 = Disagree; 3 = Neutral; 4 = Agree; 5 = Strongly Agree.

The statements used to assess technological innovation are presented in Table A1. Respondents were required to indicate their level of agreement with each statement using the five-point Likert scale provided below.

Table A1. Technological innovation

Code	Statement	1	2	3	4	5
D1	The firm continuously improves its digital platforms and systems.					
D2	Technology adoption has improved operational efficiency within the firm.					
D3	Technological innovation enhances the firm's ability to respond to market changes.					
D4	Investment in technology has improved the firm's scalability.					
D5	Technological innovation positively contributes to the firm's growth performance.					
D6	The firm is actively adopting emerging technologies such as artificial intelligence (AI) and blockchain.					
D7	The firm invests in research and development to support innovation.					

The statements used to assess organizational growth are presented in Table A2. Respondents were required to indicate their level of agreement with each statement using the five-point Likert scale provided below.

Table A2. Organizational growth

Code	Statement	1	2	3	4	5
F1	The firm's revenue has increased since market entry.					
F2	The firm's customer base has grown over time.					
F3	The firm's operational capacity has improved since its establishment.					
F4	The firm has strengthened its competitive position in the market.					
F5	The firm has experienced sustained overall growth.					
F6	The firm has expanded its product offerings or services since market entry.					
F7	The firm has entered new markets or geographic segments since establishment					